

Priyansh Pathak

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Research Summary

Research-oriented Computer Science undergraduate specializing in Artificial Intelligence and Machine Learning with strong interests in Computer Vision and Deep Learning. Experienced in deep metric learning, Siamese neural networks, and AI-driven visual recognition systems. Currently working on applied AI systems including automated attendance and intelligent route planning, with emphasis on scalable machine learning pipelines and experimental evaluation.

Education

SRM Institute of Science and Technology

Aug 2023 – May 2027

B.Tech in Computer Science and Engineering (Artificial Intelligence and Machine Learning) **CGPA: 9.2 / 10**

Indian Institute of Technology Ropar

Sept 2024 – Feb 2025

Major in Artificial Intelligence

Technical Skills

Programming: Python, Java, C++, SQL

Machine Learning: Deep Learning, CNNs, Metric Learning, Siamese Networks, Representation Learning

Computer Vision: OpenCV, MTCNN, Face Recognition, Image Processing, Video Analysis

Frameworks Tools: TensorFlow, Keras, Flask, MATLAB, Git

Libraries: NumPy, Pandas, Scikit-learn

Experience

Research Intern — DRDO CAIR, Bangalore

Jan 2026 – Present

- Conducting research on AI-based route planning and navigation optimization for defense applications.
- Implementing machine learning models to analyze spatial and environmental data for path planning.
- Studying and reproducing algorithms from recent research papers for experimental benchmarking.

Research Intern — IIT Mandi

May 2025 – Jul 2025

- Conducted research on a video-based attendance system using deep learning and computer vision.
- Implemented face detection using MTCNN and trained Siamese networks with triplet loss.
- Performed dataset preprocessing and comparative evaluation with baseline face recognition models.

Machine Learning Intern — Skybrisk Technologies

Feb 2025 – Aug 2025

- Worked on machine learning and data analysis tasks for real-world datasets.
- Developed predictive models and performed data preprocessing using Python.
- Assisted in building AI-based solutions for data-driven decision systems.

AI Research Intern— AIEnsured

Jul 2025 – Sept 2025

- Developed machine learning models with focus on fairness, robustness and explainability.
- Built evaluation pipelines to analyze bias and model performance metrics.
- Conducted experimental analysis across datasets to study model behaviour.

Computer Vision Intern— CodTech IT Solutions

Jan 2025 – Mar 2025

- Developed computer vision models for image classification tasks.
- Implemented convolutional neural networks using TensorFlow and Keras.
- Performed dataset preprocessing, augmentation and model evaluation.

Web Development Intern — GAO Tech

Sept 2024 – Dec 2024

- Developed responsive user interfaces using HTML, CSS and JavaScript.
- Implemented modular UI components and improved front-end usability.
- Collaborated with development teams to optimize application performance.

Web Design Intern — TechnoHacks EduTech

Dec 2024 – Jan 2025

- Designed responsive webpages and improved user interface layouts.
- Implemented front-end functionality using HTML, CSS and JavaScript.
- Applied modern UI/UX principles to enhance usability and accessibility.

Software Testing Intern — Codec Pvt Ltd

Dec 2024 – Jan 2025

- Performed functional testing and validation of application modules.
- Designed and executed test cases to evaluate software reliability.
- Documented defects and collaborated with developers to improve software quality.

Selected Projects

Video-Based Face Recognition Attendance System

- Developed a deep learning system using Siamese neural networks and triplet loss.
- Implemented face detection using MTCNN and facial embedding learning.
- Automated attendance logging from classroom video streams.

Real-Time Face Recognition Attendance System

- Built a Python GUI application using Tkinter and OpenCV.
- Implemented real-time face detection and recognition pipeline.
- Integrated CSV-based attendance storage and model training interface.

Skin Cancer Detection Using Deep Learning

- Developed a CNN-based model to detect skin cancer from dermoscopic images.
- Performed image preprocessing and dataset augmentation.
- Trained and evaluated classification models using TensorFlow.

Face Mask Detection

- Implemented a computer vision system using OpenCV and deep learning.
- Trained a CNN model to detect masked and unmasked faces.
- Achieved real-time detection using webcam video streams.

Movie Recommendation System

- Built a machine learning recommendation system using user preference data.
- Implemented collaborative filtering and content-based filtering approaches.
- Generated personalized movie recommendations based on genres and ratings.

Online Quiz System

- Developed a Flask-based web application for online examinations.
- Implemented authentication, question management and automated evaluation.
- Designed a responsive frontend using HTML, CSS and JavaScript.

Research Work in Progress

Deep Metric Learning for Video-Based Attendance

- Investigating Siamese neural networks with triplet loss for learning robust facial embeddings.
- Implementing a deep learning pipeline integrating MTCNN for face detection and alignment.
- Evaluating model performance on classroom video datasets for automated attendance recognition.

AI and GIS-Based Route Planning

- Researching machine learning approaches for intelligent route planning and navigation optimization.
- Studying graph-based algorithms and reinforcement learning techniques for path planning.
- Implementing prototype models to evaluate efficiency under real-world environmental constraints.

Leadership

Associate Lead — SRMKZILLA

2023 – Present

Contributor to open-source initiatives and technical community events at SRM Institute of Science and Technology.